

EE-102 Electric Circuit Analysis

Experiment # 1

Introduction to Lab Equipment

Objective/s:

- Oscilloscope
- Function Generator
- Multimeter

1. Function generator

We can get different types of wave shapes from a function generator. A function generator has a frequency range, and we can get varying amplitude from its main output. We can verify its frequency and wave shapes by applying its output to oscilloscope.

For using a function generator, we plug its probe in any of the output jack 20 V_{p-p} or 15V_{p-p}.select the required frequency and amplitude from there select switches. Moreover, different wave shapes are available. The wave shape can be observed on the oscilloscope and can be easily adjusted.

2. Oscilloscope

An oscilloscope can display and measure many electric quantities. It can measure:

1. AC/DC voltages
2. Frequency, time-period.
3. Phase difference and many more.

3. Voltage measurement

1. Connect the signal to be measured to the input jack and select CH1/CH2 to be used.
2. Set the VOLTS/DIV and TIME/DIV controls to obtain a waveform.
3. Adjust the reference level of the trace by POS control. make sure to calibrate the oscilloscope,
4. Set the AC-GND-DC switch to DC or AC.
5. use the X POS and VOLTS/DIV control to set the waveform

6. Multiply the vertical measured distance (no. of large divisions on oscilloscope) with VOLTS/DIV and probe magnification.
7. Similarly, V_{p-p} can be measured for AC voltages by counting the no. of division.

4. Tasks

1. Observations for Resistance

S No	BY COLOR CODE	READING BY D.M.M	DIFFERENCE
1			
2			
3			

2. Observation of Voltage

S No	DC Voltmeter	READING BY D.M.M	DIFFERENCE
1			
2			
3			